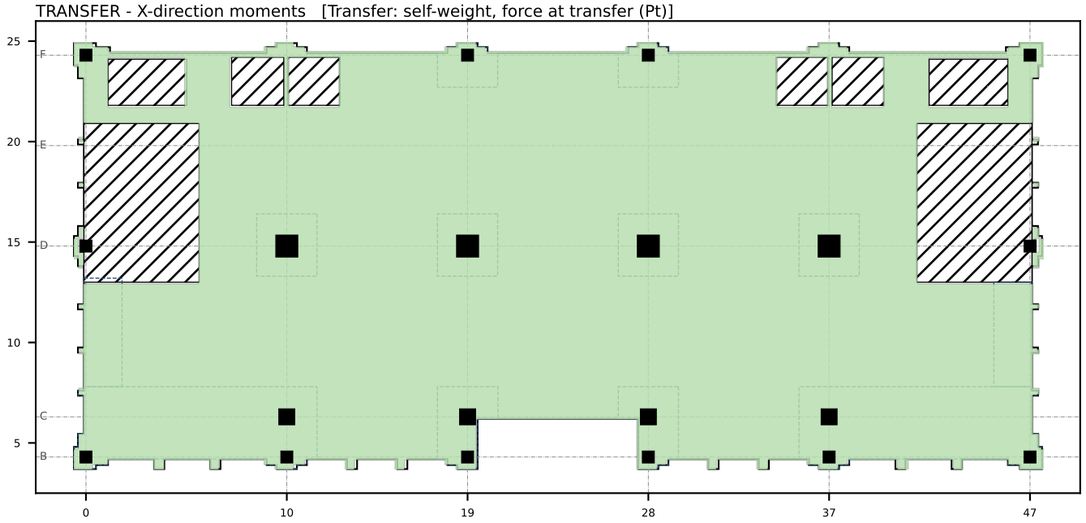
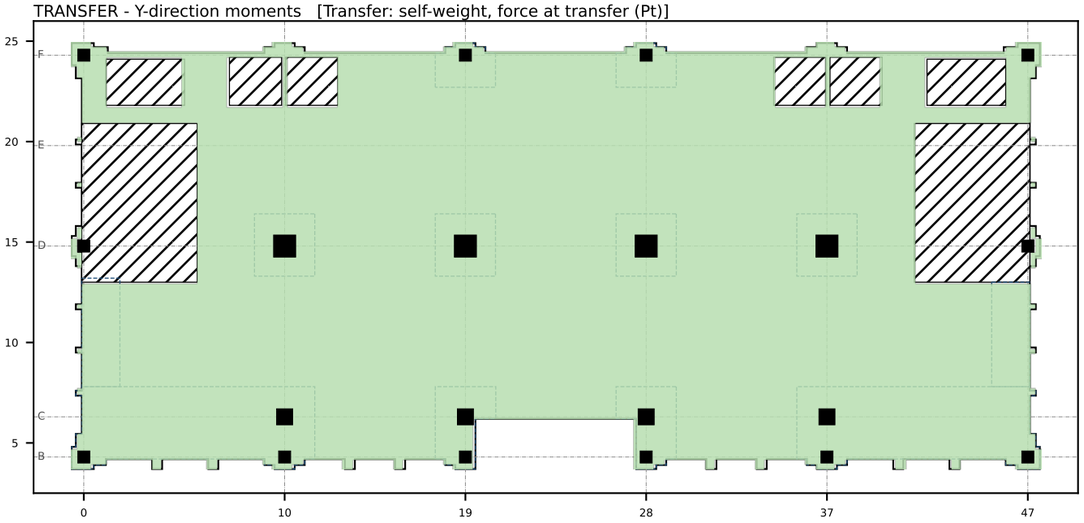
2F PT SLAB - STRESS UTILISATION BY STAGE AND DIRECTION (REDRAFT FOR REVIEW)

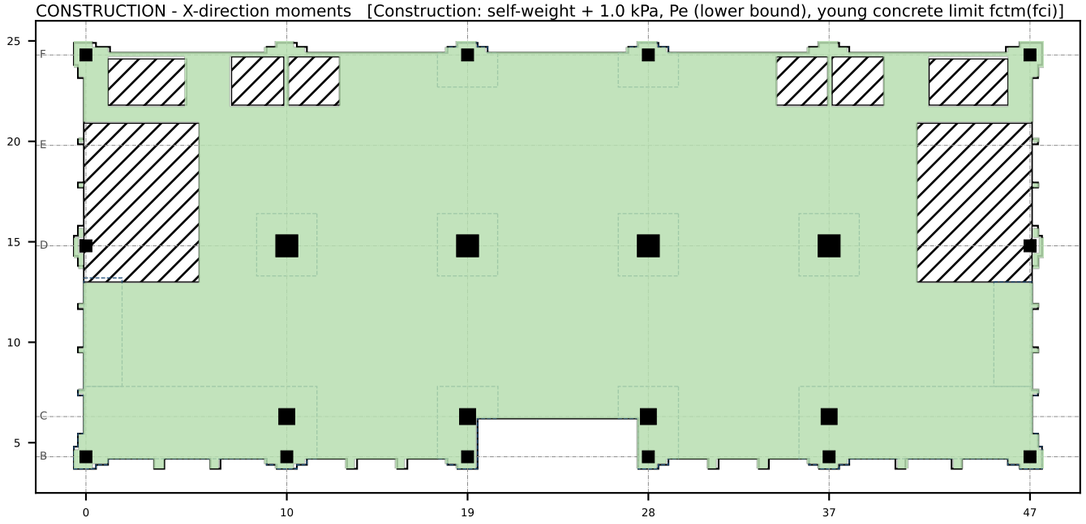
TRANSFER - X-direction moments [Transfer: self-weight, force at transfer (Pt)]



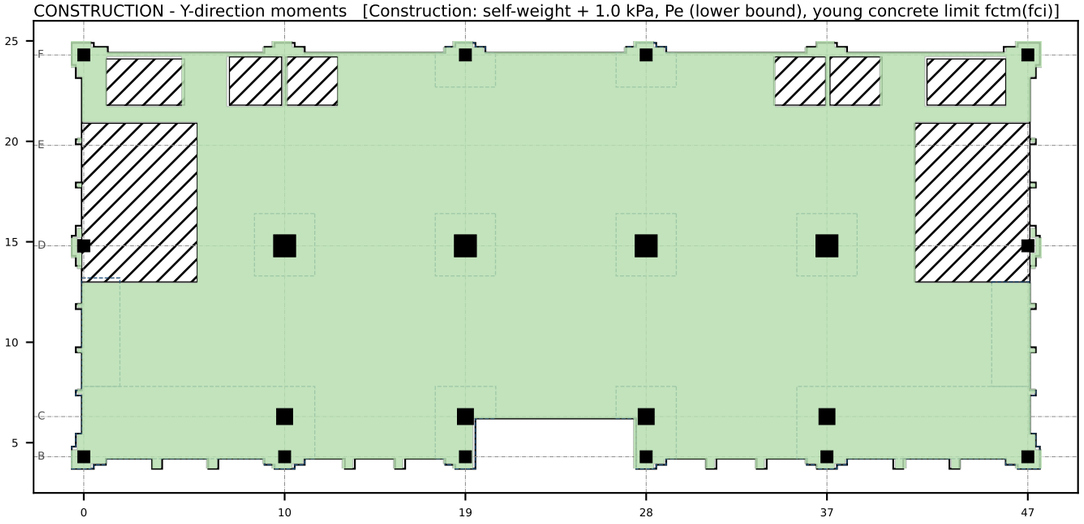
TRANSFER - Y-direction moments [Transfer: self-weight, force at transfer (Pt)]



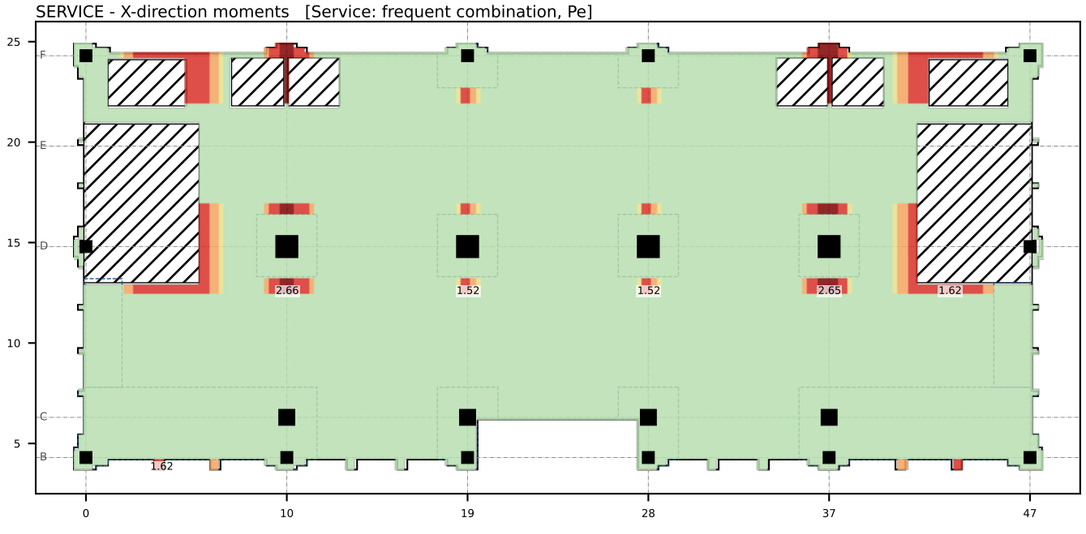
CONSTRUCTION - X-direction moments [Construction: self-weight + 1.0 kPa, Pe (lower bound), young concrete limit fctm(fci)]



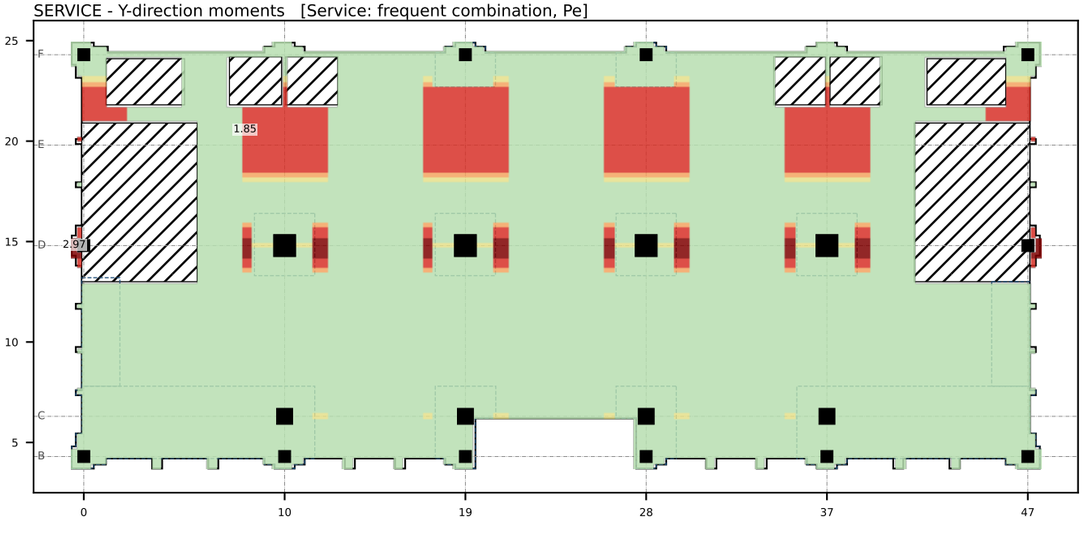
CONSTRUCTION - Y-direction moments [Construction: self-weight + 1.0 kPa, Pe (lower bound), young concrete limit fctm(fci)]

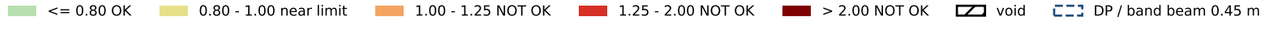


SERVICE - X-direction moments [Service: frequent combination, Pe]



SERVICE - Y-direction moments [Service: frequent combination, Pe]

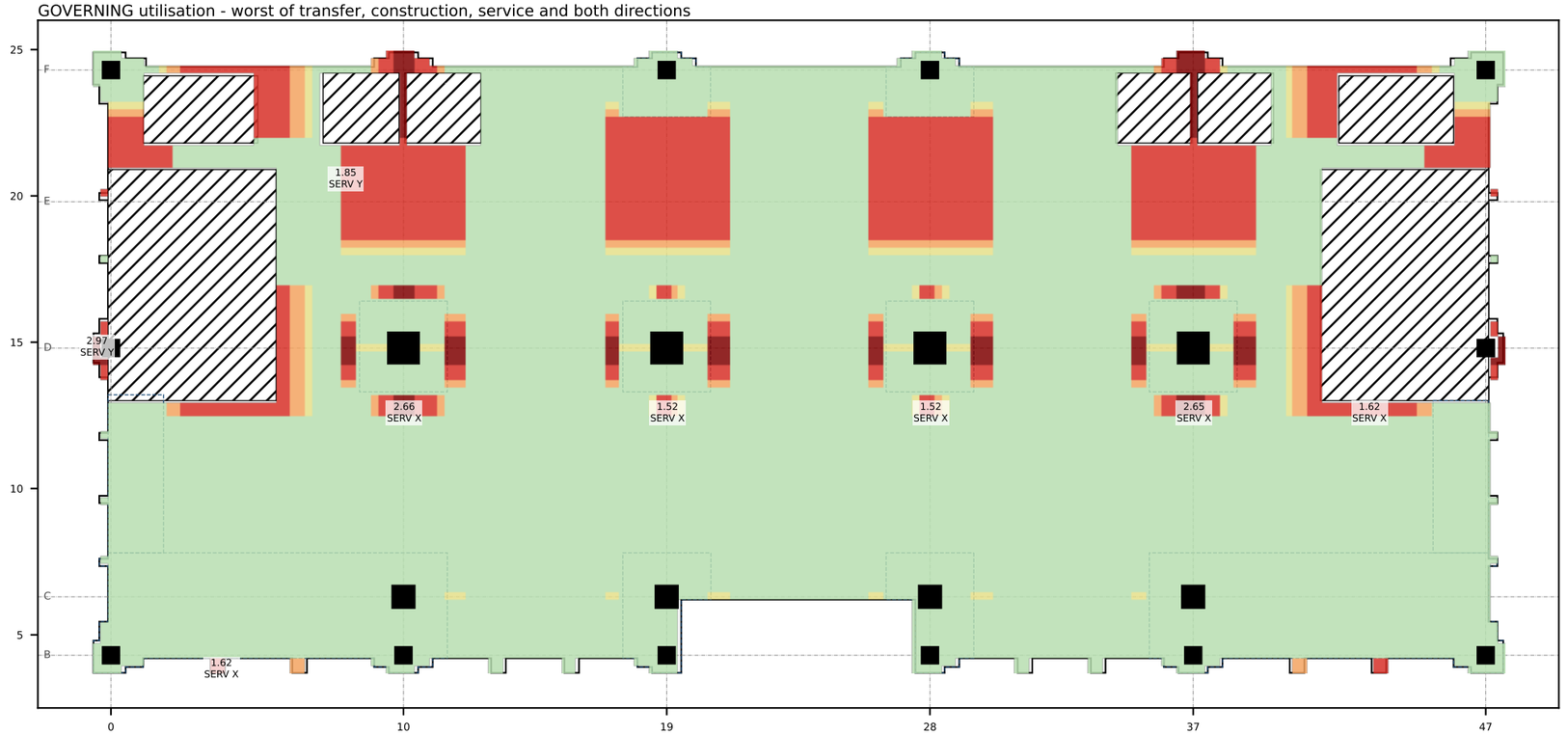




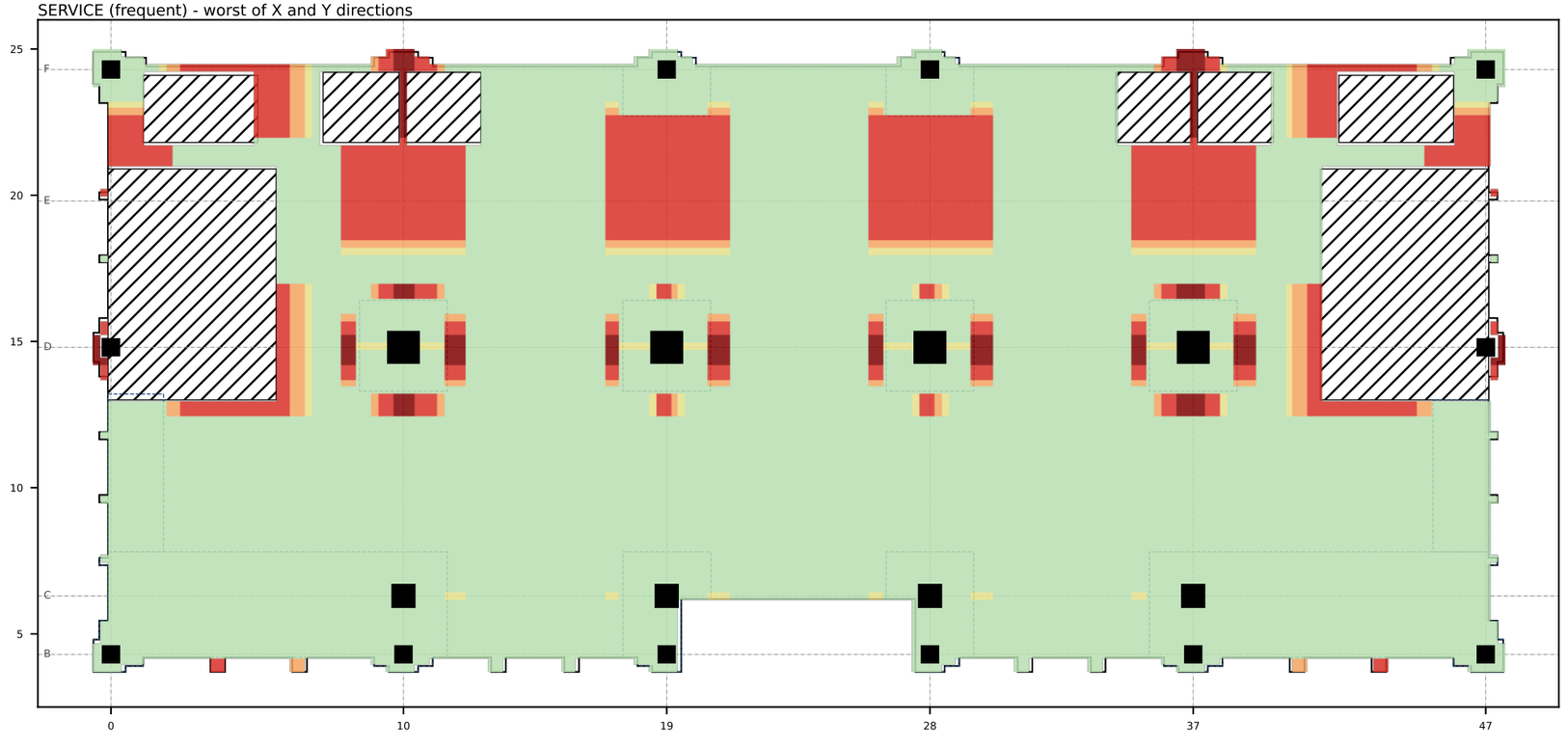
INDICATIVE SCREENING - NOT A SIGN-OFF. Utilisation = max(tension / fctm, compression / 0.6 f_c). EN 1992-1-1 Annex I column / middle strips (hog 80 %, sag 70 % in column strip); continuous 1 m strips on pinned supports; local precompression; section 0.45 m inside DP/BB else 0.25 m. PT basis ASSUMED (Gr.270, 0.75 fpu, 18 % losses); mild steel not credited. Numbers = max utilisation of a zone where > 1.00.

2F PT SLAB - WHERE THE STRIPS DO NOT PASS (REDRAFT FOR REVIEW)

GOVERNING utilisation - worst of transfer, construction, service and both directions



SERVICE (frequent) - worst of X and Y directions



<= 0.80 OK | 0.80 - 1.00 near limit | 1.00 - 1.25 NOT OK | 1.25 - 2.00 NOT OK | > 2.00 NOT OK | void | DP / band beam 0.45 m

INDICATIVE SCREENING - NOT A SIGN-OFF. Utilisation = max(tension / fctm, compression / 0.6 f_c). EN 1992-1-1 Annex I column / middle strips (hog 80 %, sag 70 % in column strip); continuous 1 m strips on pinned supports; local precompression; section 0.45 m inside DP/BB else 0.25 m. PT basis ASSUMED (Gr.270, 0.75 fpu, 18 % losses); mild steel not credited. Numbers = max utilisation of a zone where > 1.00.

ZONES / STRIPS NOT PASSING (utilisation > 1.00) - INDICATIVE, REDRAFT FOR REVIEW

Dir	Strip	Location	Transfer	Construction	Service (freq.)	Worst cell (service)
Y	column strip	support at y = 14.8 m	0.11	0.67	2.97	x=-0.5, y=14.8
X	column strip	support at x = 10.0 m	0.13	0.29	2.66	x=10.0, y=12.6
X	column strip	support at x = 37.0 m	0.13	0.29	2.65	x=37.0, y=12.6
Y	column strip	midspan y 14.8 to 24.3 m	0.10	0.37	1.85	x=8.0, y=20.6
X	column strip	midspan x 0.0 to 10.0 m	0.17	0.19	1.62	x=3.8, y=3.8
X	column strip	midspan x 37.0 to 47.0 m	0.17	0.19	1.62	x=43.0, y=12.6
X	column strip	support at x = 28.0 m	0.11	0.14	1.52	x=28.0, y=12.6
X	column strip	support at x = 19.0 m	0.11	0.14	1.52	x=19.0, y=12.6

Zones: S = within 0.15 x span of a support line (hogging); M = midspan. Strip: EN 1992-1-1 Annex I column / middle strip. Transfer and construction pass in all zones (max utilisation 0.17 and 0.67). Full list: strip_zone_utilisation.csv. NOT a sign-off.